

Ryan Westfall

✉ ryan@reflashed.com | 🏠 Ryan-Westfall.info | 💻 github.com/Ryan-Westfall | 🔗 linkedin.com/in/Ryan-Westfall/

Experience

Meta

Seattle, WA

SOFTWARE ENGINEER

May 2025 – Present

- Launched ML-based optimization models leveraging historical payment behavior signals to predict the optimal timing of enqueueing different billing events, improving payment conversion rates and booking \$29.7M in incremental revenue
- Built integrity workflows leveraging internal machine learning models to analyze user voice and chat signals, enqueue decisions into moderation and enforcement pipelines, and route events to downstream systems for automated handling
- Led end-to-end implementation of the Speaking Indicator feature across VR and Mobile platforms, building new services from the ground up, ensuring cross-platform feature consistency; launched to 200k+ MAU, logging 30,000+ interactions daily, improving user speaking observability

Amazon

Seattle, WA

SOFTWARE ENGINEER

June 2022 – May 2025

- Led development of the Targeted Temporary Price Reduction execution system, increasing promotion execution rates from 60% to 70%, delivering on VP-level goals through innovative in-memory bulk upload solution
- Architected and implemented supply-aware inventory system enabling \$9.1MM in annualized savings by preventing negative inventory promotions; solution has been leveraged by 8 different projects across platform
- Demonstrated technical leadership in AWS services including Lambda, CloudSearch, S3, Aurora DB, as well as proprietary tools while managing critical systems handling 20K promotions and \$2.5bn in promotional spend
- Led an AWS service migration for our org-level notification platform by implementing bulk email processing, event subscription management, and comprehensive monitoring resulting in successful adoption of it by multiple critical promotion related services
- Integrated internal machine learning-driven optimization models into pricing and inventory workflows, enabling data-driven promotion scheduling based on user-defined objectives (e.g., price optimization, inventory liquidation, demand shaping)

Paycom

Oklahoma City, OK

SOFTWARE DEVELOPMENT INTERN

May 2021 – August 2021

- Created full-stack messaging bot application that interfaces with the QA department's E2E testing suite allowing for asynchronous test executions: built with React app, ASP.NET REST API and a MySQL Database

University of Oklahoma

Norman, OK

RESEARCH ASSISTANT

January 2020 – May 2021

- Conducted compiler performance optimization experiment, benchmarked control and experiment groups using polyhedral model compilers PPCG and PLUTO utilizing the polyhedral benchmark suite PolyBench/C

Pixorize

New York, NY

SOFTWARE INTERN

May 2019 – July 2019

- Created an in-house text-editor application that streamlined Pixorize's content creation and management process: built a React app and Express REST API, containerized front-end and back-end using Docker, utilized E2E and unit tests

Education

University of Oklahoma

Norman, OK

B.S. WITH HONORS IN COMPUTER SCIENCE, MINOR IN MATHEMATICS - GPA: 3.95/4

August 2018 – May 2022

Personal Projects

Automatic Generated AI Content System

- Created a fleet of social media channels that automatically upload daily AI generated content
- Utilizes a host of AI services like MidJourney, ElevenLabs and, ChatGPT to generate video assets. Programmatically creates videos via React component templating library Remotion. Uses Puppeteer to automate video uploading to all the popular social media platforms

Simple Linear Regression Powerlifting Analysis

- Conducted an analysis of how a powerlifter's body weight impacts their lifts in the bench, the deadlift, and the squat
- Used Simple Linear Regression and ran checks for linearity, normality of errors, equal variances, and independence of errors

Skills

Typescript, Javascript, Java, Python, C++, C, C#, Hack, R, iOS, Swift, Android, Kotlin, HTML, CSS, React, React Native, Redux, Jest, Cypress, Webpack, Next.js, Gradle, AWS, Docker, Git, CI/CD, REST, GraphQL, SQL, PostgreSQL, MySQL, NoSQL, Amazon DynamoDB, Redis, VR/AR, AI, Machine Learning, Mobile Development, Distributed Systems, Event-Driven Architecture, Real-Time Communication, Microservices, Real-Time Systems, System Design, Kanban, Agile, Full-stack, Scalability, Monitoring, Alerting, A/B Testing, Reliability, Integrity